

lecture 03

See

sampling distribution; parametric bootstrap; information matrix; delta method; coverage

review

an example exam question

an easy one

```
1 ll_fn <- function(theta, y) {
2   alpha <- theta[1]
3   beta <- theta[2]
4   ll <- sum(dbeta(y, shape1 = alpha, shape2 = beta, log = TRUE))
5   return(ll)
6 }
7
8 est_beta <- function(y) {
9   est <- optim(par = c(2, 2), fn = ll_fn, y = y,
10              control = list(fnscale = -1),
11              method = "BFGS") # for >1d problems
12   if (est$convergence != 0) print("Model did not converge!")
13   res <- list(est = est$par)
14   return(res)
15 }
16
17 fit <- est_beta(y)
```

1. On line 4, what does `log = TRUE` do?
2. On line 10, what does `control = list(fnscale = -1)` do?
3. Describe the function `est_beta()`, both (i) the arguments it takes and (ii) the results it returns.
4. Write R code to compute $\hat{\mu} = \frac{\hat{\alpha}}{\hat{\alpha} + \hat{\beta}}$ after running the code above.

a more tedious one

Suppose we collect N random samples $y = \{y_1, y_2, \dots, y_N\}$ and model these data with a Poisson distribution with pmf $f(y_i; \lambda) = \frac{e^{-\lambda} \lambda^{y_i}}{y_i!}$ for $y_i \in \{0, 1, 2, \dots\}$ and $\lambda > 0$. Find the ML estimator of λ .

MLE Recipe

1. Write down the likelihood
2. Take log of the likelihood.
3. Simplify the log-likelihood.
4. Take derivative of the log-likelihood
5. Set derivative equal to zero; set $\lambda = \hat{\lambda}$.
6. Solve for $\hat{\lambda}$.
7. (Check second-order conditions.)

Can you sketch the process?

Do you remember the solution?

$$\hat{\lambda} = \text{avg}(y)$$

sampling distribution

We have an estimator $\hat{\theta}$.

This estimator is a **random variable**

because $\hat{\theta}$ is a function of y ,
which is random *in some way*.

random sampling

randomization

probability model

If $\hat{\theta}$ is a random variable,
then it has a distribution.

This distribution is called the
sampling distribution.

The **sampling distribution**
is the **most important**
statistical concept (by far).

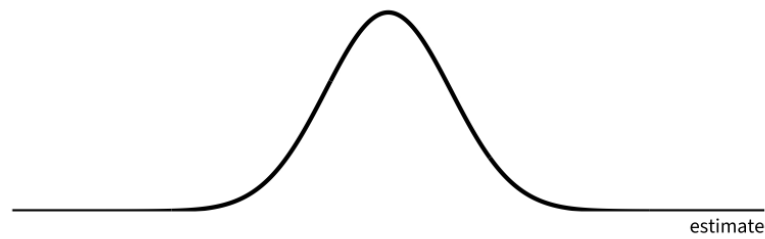
When you run a study, you get an estimate $\hat{\theta}$.
The estimate is random—it comes from a noisy procedure.
We must figure out that noise (SE, p-value, CI).

Example

The Poisson ML Estimate

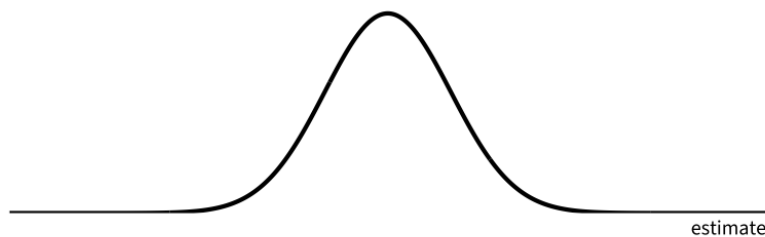
<https://gist.github.com/carlislerainey/ae1e37fe392f69017a4ce5909c4bbd29>

three important features of the sampling distribution



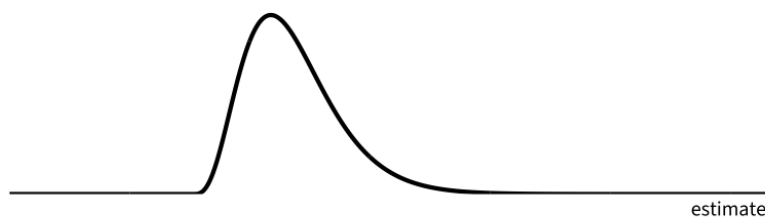
location

approx. unbiased



spread

this is the question



shape

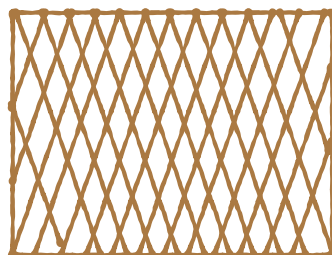
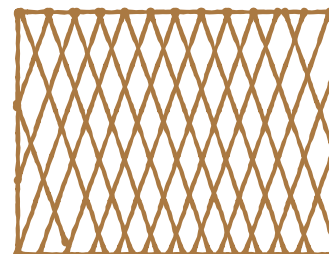
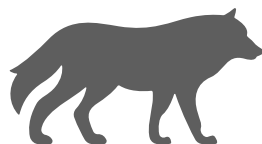
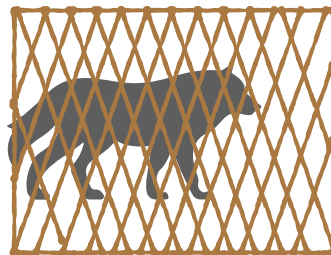
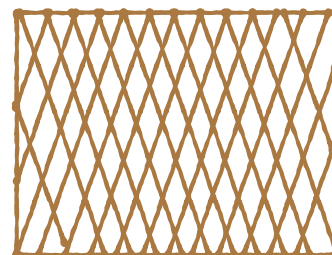
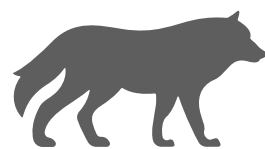
approx. normal

**How do we estimate the spread
of the sampling distribution?**



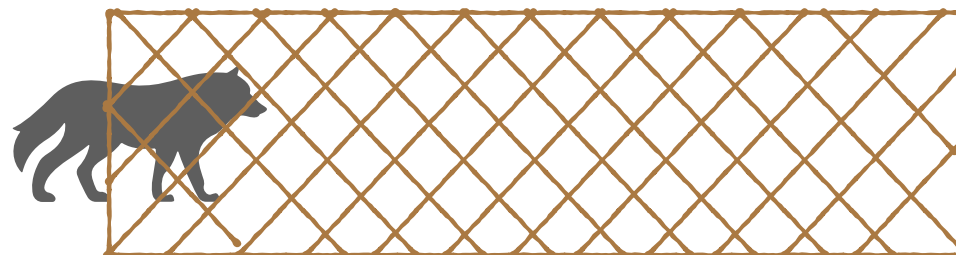
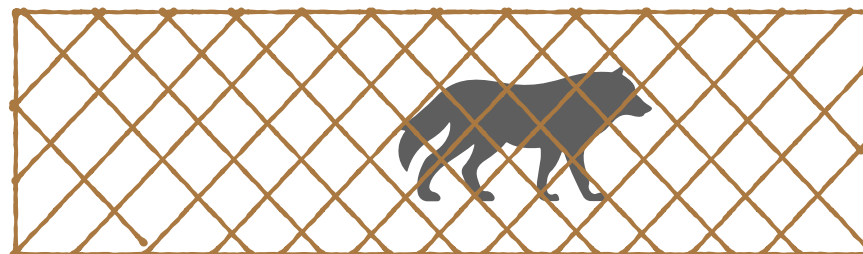
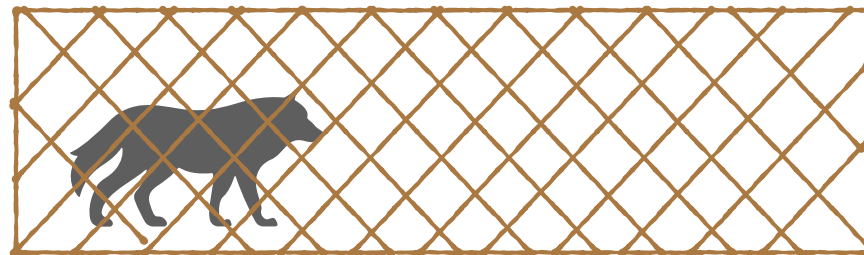
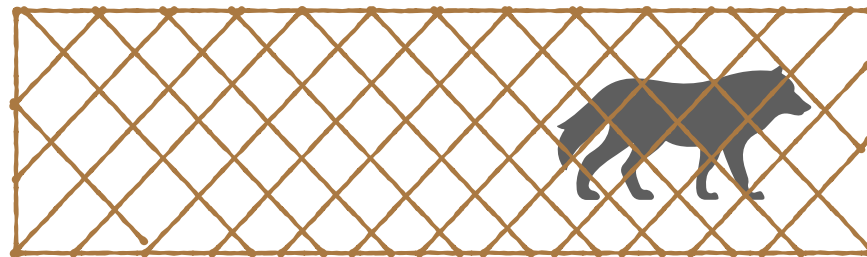


Too Narrow



Just Right

miss left about 2.5% of the time, miss right about 2.5% of the time
“almost always” catch the wolf



**So how wide does our
net need to be?**

90% CI = estimate $\pm$ 1.64 SE

95% CI = estimate $\pm$ 1.96 SE

**How do we estimate the spread
of the sampling distribution?**

parametric bootstrap

(building our intuition)

Inputs: observed data, a parametric model $f(\theta)$, an estimator $\hat{\theta}$, and perhaps a quantity of interest $\hat{\tau} = \tau(\hat{\theta})$.

Compute $\hat{\theta}$ from the observed data.

Do the following 2,000 times:

1. Simulate a new **b.s. data set** *from the fitted parametric model*.
2. Use the b.s. data set to obtain a **b.s. estimate** your quantity of interest. Store the estimate.

Summarize the empirical distribution of the b.s. estimates. (e.g., SD for SE, quantiles for CI)

Example

Toothpaste Cap Problem

<https://gist.github.com/carlislerainey/edb70a6f0da2d27f1116cc2b7606330d>

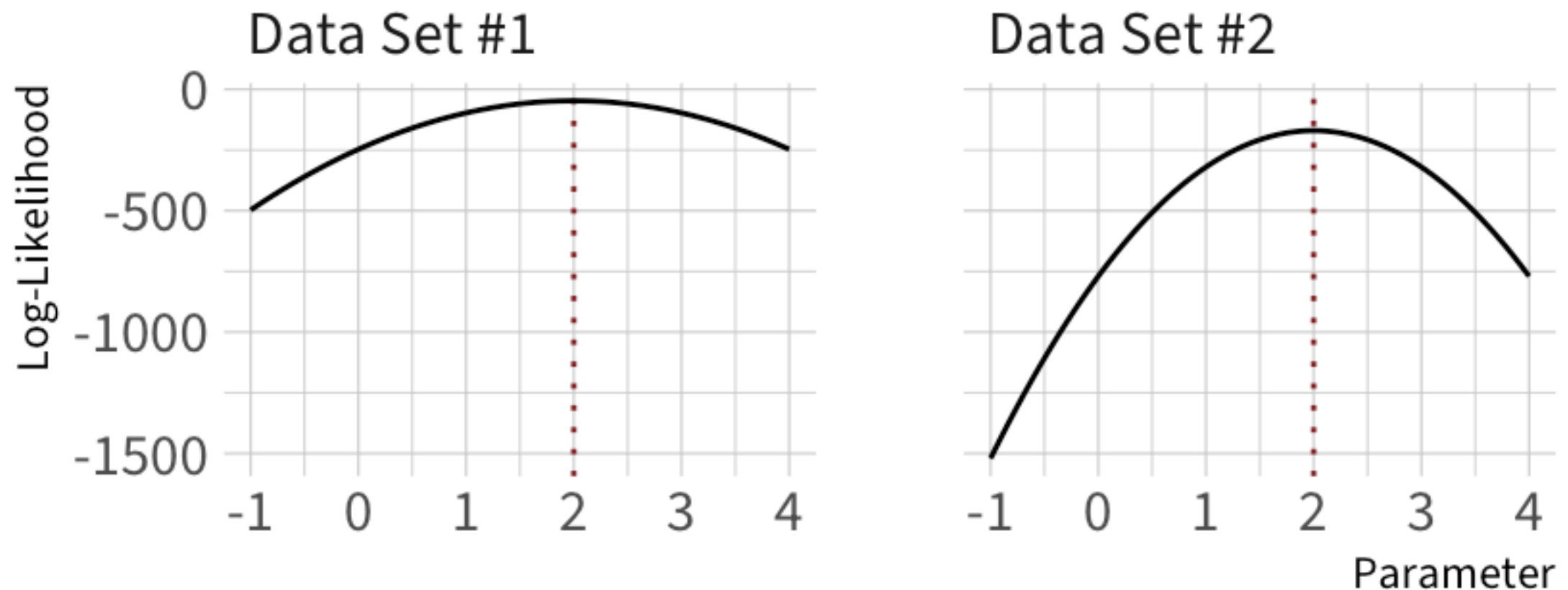
But is there a faster way?

(like a closed-form expression, maybe?)

standard errors

(a shot-in-the-dark guess)

Here are two log-likelihoods. Which produces a more precise estimate?



These are log-likelihoods from a **stylized normal model**.
(my term)

The stylized normal has unknown mean μ and known SD $\sigma = 1$.

usual normal distribution

$$f(y_i; \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2} (y_i - \mu)^2\right)$$

stylized normal distribution

$$f(y_i; \mu) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2} (y_i - \mu)^2\right)$$

(keep going on board; see handout)

standard errors

(for single parameter models)

just a magical result

the inverse of the negative curvature
is a great estimate of the variance

(this is for the *log*-likelihood)

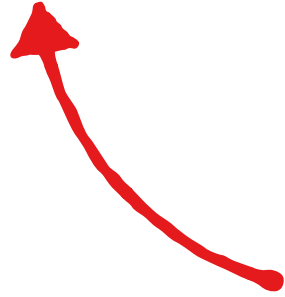
$$\left[-\ell''(\hat{\theta}) \right]^{-1}$$

inverse of the negative curvature: “INC”

asymptotic results

Definition (Fisher Information). For a model $f(y; \theta)$ with log-likelihood $\ell(\theta) = \sum_{i=1}^N \log f(y_i; \theta)$,

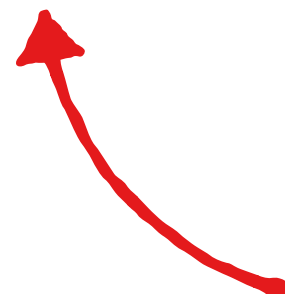
- the *expected information* is defined as $\mathcal{I}(\theta) = -\mathbb{E}_{\theta} [\nabla^2 \ell(\theta)]$ and
- the *observed information* is given by the negative Hessian of the log-likelihood at the ML estimate, $\mathcal{I}_{\text{obs}}(\hat{\theta}) = -\nabla^2 \ell(\hat{\theta})$.



just NC (no I)

Theorem (Asymptotic Normality of ML Estimators) Let $y_1, \dots, y_N$ be iid from a model $f(y; \theta)$ with true parameter θ . Suppose the regularity conditions for consistency hold and the Fisher information matrix $\mathcal{I}(\theta)$ is finite and positive definite. Then the maximum likelihood estimator $\hat{\theta}$ is asymptotically normal so that

$$\hat{\theta} \stackrel{a}{\sim} \mathcal{N}(\theta, \mathcal{I}(\theta)^{-1}).$$



just INC

location

spread

$$\hat{\theta} \stackrel{a}{\sim} N(\theta, I(\theta)^{-1})$$

shape

just INC

The diagram shows the equation $\hat{\theta} \stackrel{a}{\sim} N(\theta, I(\theta)^{-1})$. A green arrow points from the word 'location' to the parameter θ . A red arrow points from the word 'spread' to the covariance matrix $I(\theta)^{-1}$, which is enclosed in a red rectangular box. A blue arrow points from the word 'shape' to the normal distribution symbol N . A red curved arrow points from the text 'just INC' to the red box around $I(\theta)^{-1}$.

Theorem (Asymptotic Variance of ML Estimators). Under the conditions of the asymptotic normality theorem, the covariance of the maximum likelihood estimator is well-approximated by the inverse Fisher information

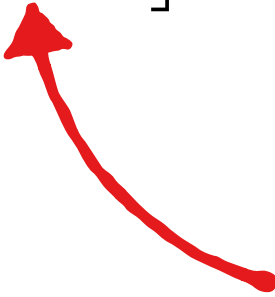
$$\text{Var}(\hat{\theta}) \approx \mathcal{I}(\theta)^{-1}.$$

In practice, we replace the unknown θ with the maximum likelihood estimate and use the observed information:

$$\widehat{\text{Var}}(\hat{\theta}) \approx \left[-\nabla^2 \ell(\hat{\theta}) \right]^{-1}.$$

For a single parameter θ , this reduces to

$$\widehat{\text{Var}}(\hat{\theta}) \approx \left[-\frac{\partial^2 \ell(\theta)}{\partial \theta^2} \Big|_{\theta=\hat{\theta}} \right]^{-1} = \left[-\ell''(\hat{\theta}) \right]^{-1}.$$



just INC

Example

Toothpaste Cap Problem

(on board; see handout)

delta method

In the one-parameter case, $\widehat{\text{Var}}[\tau(\hat{\theta})] \approx \left(\tau'(\hat{\theta})\right)^2 \cdot \widehat{\text{Var}}(\hat{\theta})$.

$$\text{Or, } \widehat{\text{SE}}[\tau(\hat{\theta})] \approx |\tau'(\hat{\theta})| \cdot \widehat{\text{SE}}(\hat{\theta}).$$

Let's call $\tau'(\hat{\theta})$ the “slope”—its just an adjustment factor.

Simple Example to Build Intuition: Convert from π as a Proportion to a Percent

$$\text{Var}(\hat{\pi}) = \frac{\hat{\pi}(1 - \hat{\pi})}{N} \text{ is already established.}$$

$\text{pct} = \tau(\pi) = 100 \times \pi$ is the desired transformation.

$\tau'(\pi) = 100$, because of the constant rule $[ax]' = a$.

$$\text{Then } \left(\tau'(\hat{\theta}) \right)^2 \times \widehat{\text{Var}}(\hat{\theta}) \text{ is } (100)^2 \times \left[\frac{\hat{\pi}(1 - \hat{\pi})}{N} \right] \text{ or } 10,000 \times \left[\frac{\hat{\pi}(1 - \hat{\pi})}{N} \right].$$

$$\text{Then SE} = \sqrt{10,000 \times \left[\frac{\hat{\pi}(1 - \hat{\pi})}{N} \right]} = 100 \times \sqrt{\frac{\hat{\pi}(1 - \hat{\pi})}{N}}.$$

“ π is about 0.053 g.o.t. 0.018 or so”

becomes “the percent is about 5.3 g.o.t. 1.8 or so.”

A Slightly More Complex Example to Build Intuition: Square a Parameter

$\text{Var}(\hat{\theta})$ is something—doesn't matter what

$\tau(\theta) = \theta^2$ is the desired transformation (e.g., SD to variance).

$\tau'(\theta) = 2\theta$, because of the power rule.

Then $\left(\tau'(\hat{\theta})\right)^2 \times \widehat{\text{Var}}(\hat{\theta})$ is $\left[2\hat{\theta}\right]^2 \times \text{Var}(\hat{\theta})$.

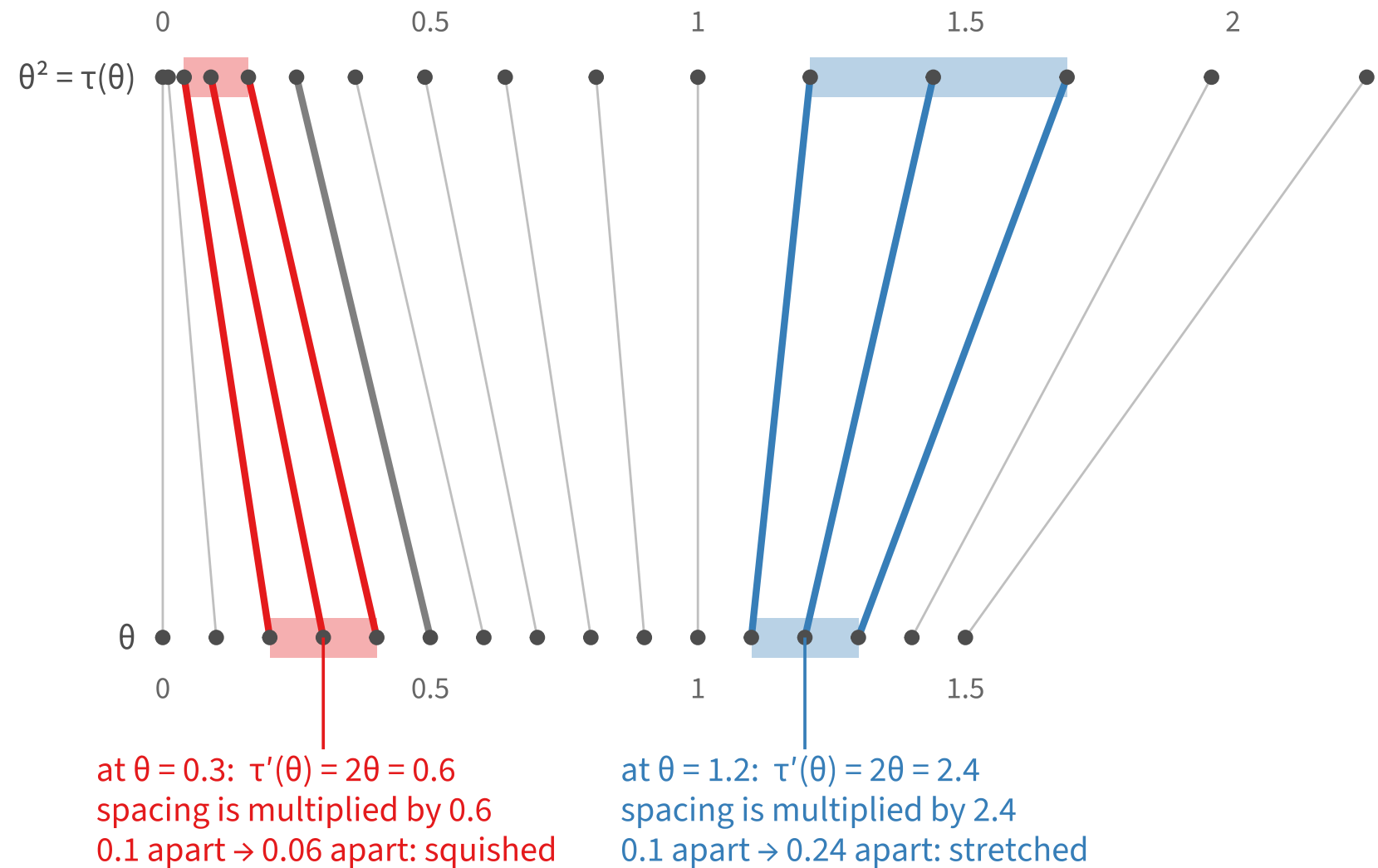
Then $\text{SE}(\hat{\theta}) = 2\hat{\theta} \times \sqrt{\text{Var}(\hat{\theta})}$.

Notice that the slope or adjustment factor is $2\hat{\theta}$.

The delta method illustrated

How the transformation might stretch or squish the space.

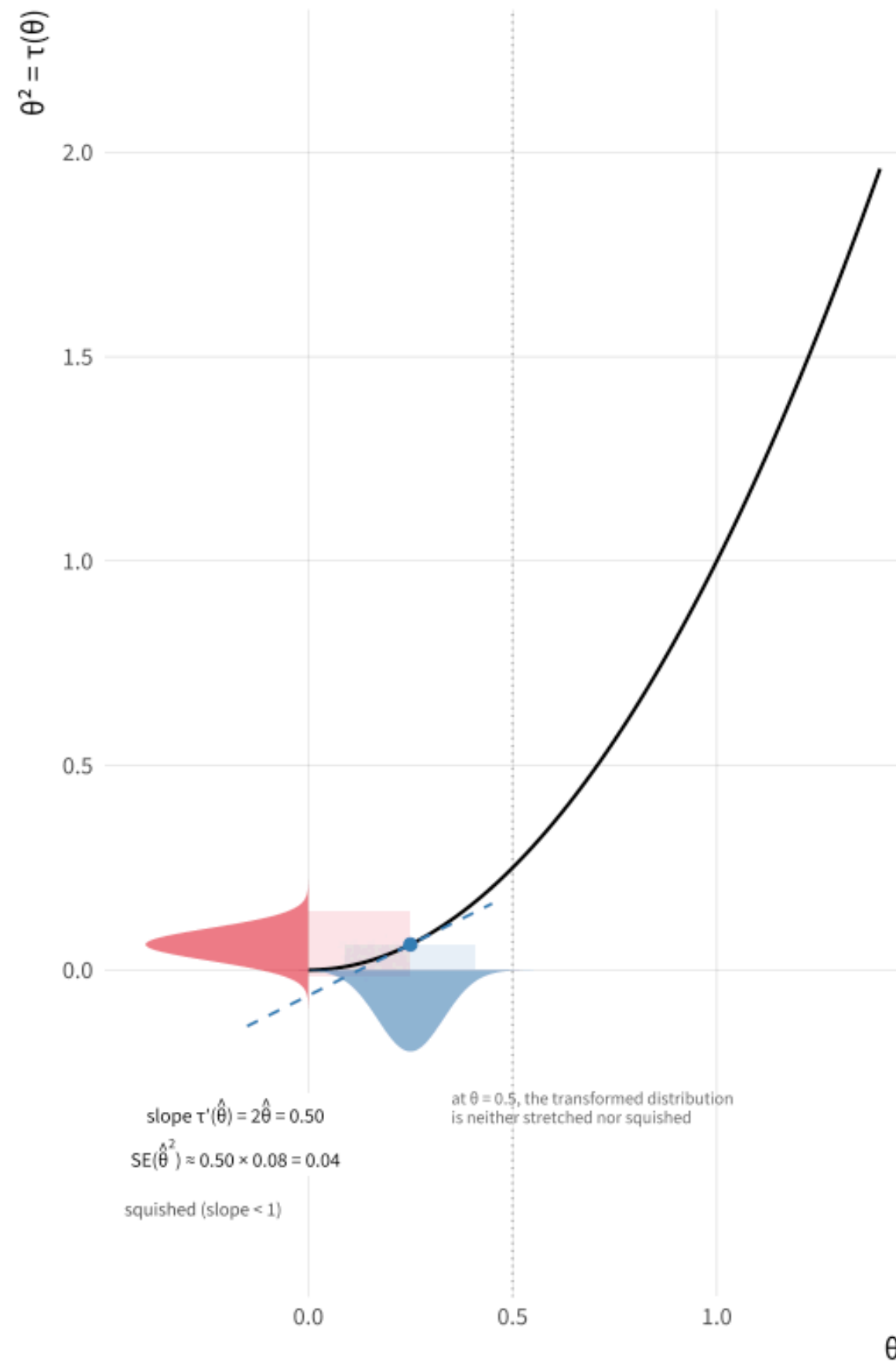
at $\theta = 0.5$, the transformed values are neither stretched nor squished



In general: $SE(\hat{\theta}^2) \approx \tau'(\hat{\theta}) \cdot SE(\hat{\theta})$ and $Var(\hat{\theta}^2) \approx [\tau'(\hat{\theta})]^2 \cdot Var(\hat{\theta})$

The delta method illustrated

How the transformation might stretch or squish the sampling distribution.



Example

Toothpaste Cap Problem

(on board; see handout)

standard errors

(for multi-parameter models)

replace the scalars with their matrix equivalents

$$\frac{\partial^2 \ell(\theta)}{\partial \theta^2} \text{ becomes } \nabla^2 \ell(\theta)$$

(second derivative becomes the Hessian)

$$\widehat{\text{Var}}[\tau(\hat{\theta})] \approx \left(\tau'(\hat{\theta}) \right)^2 \cdot \widehat{\text{Var}}(\hat{\theta})$$

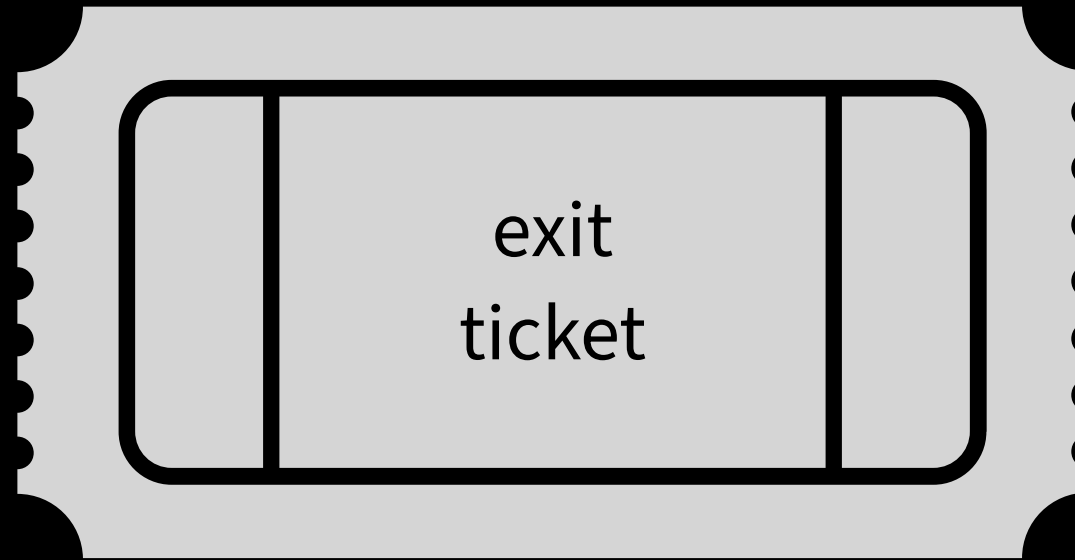
$$\text{becomes } \widehat{\text{Var}}(\tau(\hat{\theta})) \approx \nabla \tau(\hat{\theta})^\top \cdot \widehat{\text{Var}}(\hat{\theta}) \cdot \nabla \tau(\hat{\theta})$$

(derivative becomes gradient; square becomes matrix sandwich analog)

Example

Normal Model of NOMINATE w/ `optim()`

<https://gist.github.com/carlislerainey/60d47883a470687f10d5f632e243ddaf>



What's one new idea from today that
changed the way you understand
something from before?